Programming with Data Midterm Project

World Happiness Report 2019 Analysis

Introduction

Happiness is a universal pursuit, and the understanding of its determinants has been a concern of considerable research effort within economics, sociology, and psychology. This project explores the reasons that account for variation in happiness across the globe, using data from international reports and indices. In the project, a machine learning approach is used to build a predictive model that estimates happiness scores by using socio-economic and health indicators such as GDP per capita, social support, and healthy life expectancy.

The research is designed to derive useful insights from the dataset and present the findings in a series of visualizations, statistical analysis, and machine learning predictions. This project showcases not only the impact different variables have on happiness but also how machine learning techniques can be used in real-world social science problems.

Aims and Objectives

Aim

The primary aim of this project is to understand the factors contributing to happiness and analyze their relationships with happiness scores across various countries. This study takes an exploratory approach through data analysis and predictive modeling, generating findings that could inform future research and policy decisions.

Objectives

1. Enhanced Data Exploration

Conduct a detailed analysis of the dataset to reveal patterns, trends, and correlations.

2. Advanced Predictive Modeling

Develop a robust regression-based model to estimate happiness scores using selected indicators.

Methodology

Data Acquistion

The dataset for this project is the "World Happiness Report 2019," sourced from Kaggle. It provides a comprehensive view of happiness levels across countries, along with socio-economic and health-related metrics. The dataset is well-suited for analysis and predictive modeling due to its reliability, clarity, and structured format.

Exploratory Data Analysis (EDA)

Data Cleaning

Address missing values, inconsistencies, and outliers to ensure the dataset is clean and ready for analysis. This step ensures the integrity and reliability of the results.

Visualisation

Create compelling visualizations to highlight key patterns, trends, and correlations in the data, making the insights accessible and intuitive.

Statistical Analysis

Use statistical methods to quantify the relationships between happiness scores and various indicators, identifying significant contributors to happiness.

Predictive Modelling

Develop and evaluate a regression-based predictive model to estimate happiness scores. The model will leverage indicators such as GDP per capita, social support, and healthy life expectancy to demonstrate practical applications of the findings.

Research Area

The research focusses on identifying the underlying factors that influence happiness scores and the extente of their impact. By analysing features such as GDP per capita, social support and life expectancy, the projects will explain how economic and social factors shape global well-being. Moreover, the predictive modeling included will add a forward-looking perspective to estimate the happiness scores for new data,showing practical applications of the findings.

Dataset

The project uses the "World Happiness Report 2019" dataset, which includes several metrics on happiness for countries of the world. The data includes indicators such as:

Score (Happiness Score):

This is the target variable that shows the overall score indicating the perceived happiness level in a country.

GDP per capita:

The economic output per person, adjusted for purchasing power.

Social support:

The presence of networks of support for the individual.

Healthy life expectancy:

Average life expectancy, adjusted for healthy quality.

Freedom to make life choices:

This refers to the perceived ability of individuals in a country to make choices regarding their own lives, such as career, education, and personal decisions.

Generosity:

This metric measures the willingness of individuals in a country to donate to charity and help others in need.

Perception of corruption:

This indicator measures how widespread corruption is perceived to be in both business and government.

Considerantion of two other datasets

The OECB Better Life Index and Gallup World Poll were taken into consideration when the project initially began. </br> </br> The Gallop World Poll depends on subjective responses, which can be swayed by cultural biases, even while it offers comprehensive, individual-level data on life satisfaction across several nations and allows for profound insights into how individuals view their own well-being. </br></br> The OECD Better Life Index, on the other hand, provides a thorough approach to quality of life by taking into account non-economic aspects like work-life balance and environmental quality. However, it gives less emphasis on subjective well-being or personal happiness and might not fully capture the emotional experience of life satisfaction.

Why this dataset was chosen?

There are several reasons I chose the World Happiness Report 2019 dataset:

1. It provides a direct measure of happiness with contributing factors, thus it is suitable for analysing relationships and building predictive models.
2. The dataset is obtained from a reliable source (Kaggle) and is publicly available for analysis.
3. It is clean organized, therefore, does not require extensive preprocessing to process and analyse.
4. Happiness is a critical indicator of societal welfare, and studying this dataset will provide actionable insights for policymakers and researchers.

Ethics of Data Use

The data has been taken from the World Happiness Report 2019, available on Kaggle. This dataset aggregates information from well-known public surveys and reports. Its methodology and sources are clearly stated, hence transparent.

Anonymity

The data are at a national level of aggregation, and no personal or sensitive information is included, which protects privacy.

Transparency

Data have been transparently gathered via public reports and surveys whose methods are described in the World Happiness Report.

Possible bias

The dataset may over-represent high-income and middle-income countries, which are expected to result in some underrepresentation of low-income countries. This is recognized as such, and generalizations should not be made from findings without caution.

Fair Use

The analysis is conducted according to the dataset's terms of use and involves no unethical practice, including unauthorized redistribution or misrepresentation of findings.

Project background

Why the field is of interest?

Happiness has always been a deeply personal and fascinating topics for me. It is an experience shared universally, yet so varied across cultures, societies and individuals. For someone interested in understanding what makes people thrive, the intersection of happiness and data analytics is incredibly exciting. The study of happiness enables us to transcend abstractions and, instead, use concrete data to identify what truly matters for well-being. Most compelling, this field speaks to a blend of psychology, sociology and economics and gives insight into human flourishing as a whole.

The idea behind using data to uncover the patterns underlying happiness resonated with me. It was based on my curiosity about human nature coupled with the technical satisfaction in problem-solving. And knowing that the research will serve to inform policy or even improve lives took it one step further.

Previous exploration of the topic

For more than a decade, the World Happiness Report has provided invaluable insights into worldwide well-being. It has been widely utilised to emphasise the significance of things such as social support, economic stability, and health. While being fascinated by the results, I've long wondered how well these criteria predict happiness. Although previous descriptive studies give useful correlations, they do not fully investigate how these factors interact to predict pleasure in a more dynamic manner.

This interest inspired me to begin on this project, where I want to go beyond correlations and use predictive modelling approaches. By focussing on specific socioeconomic metrics such as GDP per capita and social support, I believe we may gain deeper insights. This method provides for a more in-depth understanding of how various indicators impact happiness, taking into consideration not just their presence but also their interaction and contribution to well-being. This project provides a new perspective, transitioning from descriptive research to a more predictive and practical model.

Scope of work

For this project, I aim to:

1. Analyse relationships between happiness and key socio-economic indicators like GDP per capita, social support and health.
2. Build a predictive model that estimates happiness scores based on these indicators.
3. Offer insights into how societies can improve well-being, with practical applications for policymakers.

Steps in analytical data processing pipeline

1. Obtain the dataset

Download the World Happiness Report 2019 dataset from Kaggle.

2. Data Cleaning and Preparation

After obtaining the data, take a careful look at the data to address missing values, inconsistencies and prepare it for analysis. This step is to ensure the dataset is ready for further analysis and transformation.

3. Exploratory Data Analysis (EDA)

Visualizations and descriptive statistics are used to uncover key patterns and relationships within the dataset. This step helps in understanding the underlying distribution and correlations between various variables, such as social support, GDP per capita, and healthy life expectancy, which influence happiness.

4. Define Inclusion and Exlusion criteria

Inclusion: Variables like social support, GDP per capita and healthy life expectancy were included in both EDA and the predictive model since they established connections to happiness.

Exclusion: Variables like generosity and perceptions of corruption, were analysed during EDA but excluded from final predictive model since they exhibited weaker correlations with happiness score.

Additionally, variables like environmental quality and political stability were excluded due to limitied availability or inconsistent reporting across countries which could reduce the model's stability.

5. Develop Predictive Model

Develop a regression-based machine learning model to predict happiness scores.

6. Model Testing and Evaluation

The performance of the predictive model is evaluated using standard metrics such as R-squared, Mean Absolute Error (MAE), or Root Mean Squared Error (RMSE). Test the model's performance and reflecting on its strengths, limitations and potential improvements.

Evaluation of aims and objectives

This project is inspired by a deep interest in the drivers of happiness and a desire to explore how data can inform these discussions. The objectives understanding the factors behind happiness and building a predictive model align closely with this passion. By combining exploratory analysis with predictive modeling, the project not only meets its goals but also contributes a unique perspective to a topic of universal relevance.

Technical exploration of dataset

Import libraries

To conduct our analysis, we first import essential libraries, including Pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualization, and Scikit-learn for preprocessing and modeling.

```
In [1]: import pandas as pd
        import numpy as np
        import matplotlib.pyplot as plt
        import seaborn as sns
        from sklearn.preprocessing import StandardScaler
        from sklearn.linear_model import LinearRegression
        from sklearn.model_selection import train_test_split
        from sklearn.metrics import mean_squared_error, r2_score
```

Load the Dataset

Load the World Happiness Report 2019 dataset into a dataframe named 'df' for analysis.

```
In [2]: df = pd.read_csv('Kaggle - World Happiness Report 2019.csv')
```

Data exploration and initial analysis

Initial Overview

Inspect the first five rows to understand the dataset's structure and content.

```
In [3]: df.head()
```

Out[3]:

	Overall rank	Country or region	Score	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
0	1	Finland	7.769	1.340	1.587	0.986	0.596	0.153	0.393
1	2	Denmark	7.600	1.383	1.573	0.996	0.592	0.252	0.410
2	3	Norway	7.554	1.488	1.582	1.028	0.603	0.271	0.341
3	4	Iceland	7.494	1.380	1.624	1.026	0.591	0.354	0.118
4	5	Netherlands	7.488	1.396	1.522	0.999	0.557	0.322	0.298

This step reveals columns such as Score, GDP per capita, Social support, and others that contribute to happiness measurements globally.

Dataset summary

Checking for missing values.

```
In [4]: df.isnull().sum()
```

```
Out[4]: Overall rank                    0
        Country or region               0
        Score                           0
        GDP per capita                  0
        Social support                  0
        Healthy life expectancy         0
        Freedom to make life choices    0
        Generosity                      0
        Perceptions of corruption       0
        dtype: int64
```

This command shows that the dataset no missing values are found, indicating the dataset ready for analysis.

Check for out-of-bound values

Verifying that the values in the dataset fall within plausible and reasonable ranges. For instance, checking if numeric columns such as GDP per capita, social support, and healthy life expectancy have valid values (e.g., GDP should not be negative, social support should be between 0 and 2). This ensures the dataset's integrity by identifying any unusual or incorrect values that could distort the analysis.

```
In [5]: valid = 0

        # Check values in 'GDP per capita'
        if (df['GDP per capita'] >= 0).all():
            valid += 1

        # Check values in 'Social support'
        if (df['Social support'] >= 0).all() and (df['Social support'] <= 2).all():
            valid += 1

        # Check values in 'Healthy life expectancy'
        if (df['Healthy life expectancy'] >= 0).all() and (df['Healthy life expectancy'] <= 100).all():
            valid += 1

        # Check values in 'Freedom to make life choices'
        if (df['Freedom to make life choices'] >= 0).all() and (df['Freedom to make life choices'] <= 1).all():
            valid += 1

        # Check values in 'Generosity'
        if (df['Generosity'] >= 0).all() and (df['Generosity'] <= 1).all():
            valid += 1

        # Check values in 'Perceptions of corruption'
        if (df['Perceptions of corruption'] >= 0).all() and (df['Perceptions of corruption'] <= 1).all():
            valid += 1

        #
        if valid == 6:
            print("Dataframe is valid.")
        else:
            print("Invalid values in dataframe.")

        Dataframe is valid.
```

Statistical Summary

Understanding the distribution and variability of the data.

```
In [6]: df.describe()
```

Out[6]:

	Overall rank	Score	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
count	156.000000	156.000000	156.000000	156.000000	156.000000	156.000000	156.000000	156.000000
mean	78.500000	5.407096	0.905147	1.208814	0.725244	0.392571	0.184846	0.110603
std	45.177428	1.113120	0.398389	0.299191	0.242124	0.143289	0.095254	0.094538
min	1.000000	2.853000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	39.750000	4.544500	0.602750	1.055750	0.547750	0.308000	0.108750	0.047000
50%	78.500000	5.379500	0.960000	1.271500	0.789000	0.417000	0.177500	0.085500
75%	117.250000	6.184500	1.232500	1.452500	0.881750	0.507250	0.248250	0.141250
max	156.000000	7.769000	1.684000	1.624000	1.141000	0.631000	0.566000	0.453000

This step provides a statistical overview, including mean and standard deviation, helping identify data distributions and potential outliers.

Top 10%

```
In [7]: df_sorted = df.sort_values(by='Score', ascending=False)

        top_10_percent = int(len(df) * 0.1)
        top_10_df = df_sorted.head(top_10_percent)

        print("\033[1mTop 10% of Happiness Scores\033[0m")
        print(top_10_df[['Country or region', 'Score']])

        Top 10% of Happiness Scores
           Country or region  Score
        0            Finland  7.769
        1            Denmark  7.600
        2             Norway  7.554
        3            Iceland  7.494
        4        Netherlands  7.488
        5        Switzerland  7.480
        6             Sweden  7.343
        7        New Zealand  7.307
        8             Canada  7.278
        9            Austria  7.246
        10         Australia  7.228
        11        Costa Rica  7.167
        12            Israel  7.139
        13        Luxembourg  7.090
        14    United Kingdom  7.054
```

Bottom 10%

```
In [8]: bottom_10_percent = int(len(df) * 0.1)
        bottom_10_df = df_sorted.tail(bottom_10_percent)

        print("\033[1mBottom 10% of Happiness Scores\033[0m")
        print(bottom_10_df[['Country or region', 'Score']])

        Bottom 10% of Happiness Scores
                    Country or region  Score
        141                   Comoros  3.973
        142                Madagascar  3.933
        143                   Lesotho  3.802
        144                   Burundi  3.775
        145                  Zimbabwe  3.663
        146                     Haiti  3.597
        147                  Botswana  3.488
        148                     Syria  3.462
        149                    Malawi  3.410
        150                     Yemen  3.380
        151                    Rwanda  3.334
        152                  Tanzania  3.231
        153               Afghanistan  3.203
        154  Central African Republic  3.083
        155               South Sudan  2.853
```

Average GDP for the Top and Bottom 10%

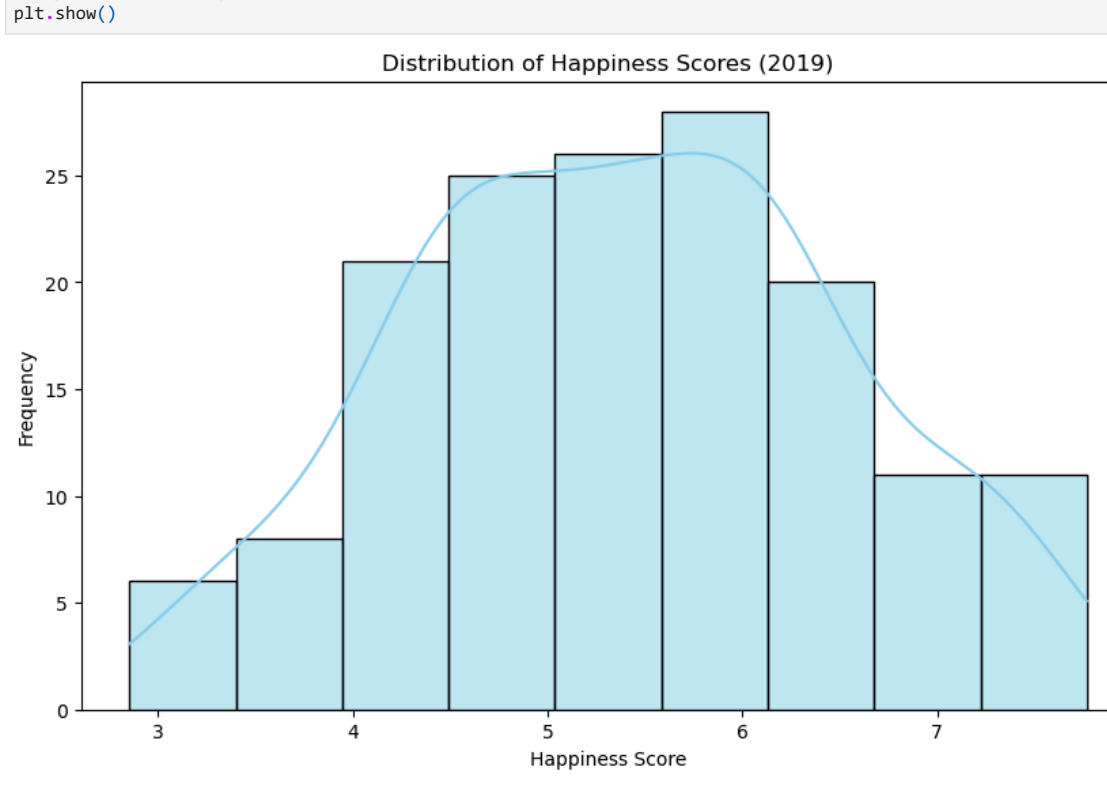
```
In [9]: top_10_avg_gdp = top_10_df['GDP per capita'].mean()
        bottom_10_avg_gdp = bottom_10_df['GDP per capita'].mean()

        print("Top 10% Average GDP:", top_10_avg_gdp)
        print("Bottom 10% Average GDP:", bottom_10_avg_gdp)

        Top 10% Average GDP: 1.3662666666666667
        Bottom 10% Average GDP: 0.36179999999999995
```

This leads us to believe that social well-being is significantly influenced by GDP. However, it also highlights the complexity of happiness, showing that social support, freedom, and a healthy life expectancy are all important factors that cannot be ensured by economic wealth alone.

Distribution of happiness scores

Explore the distribution of happiness scores.

```
In [10]: plt.figure(figsize=(10, 6))
         sns.histplot(df['Score'], kde=True, color='skyblue')
         plt.title('Distribution of Happiness Scores (2019)')
         plt.xlabel('Happiness Score')
         plt.ylabel('Frequency')
         plt.show()
```

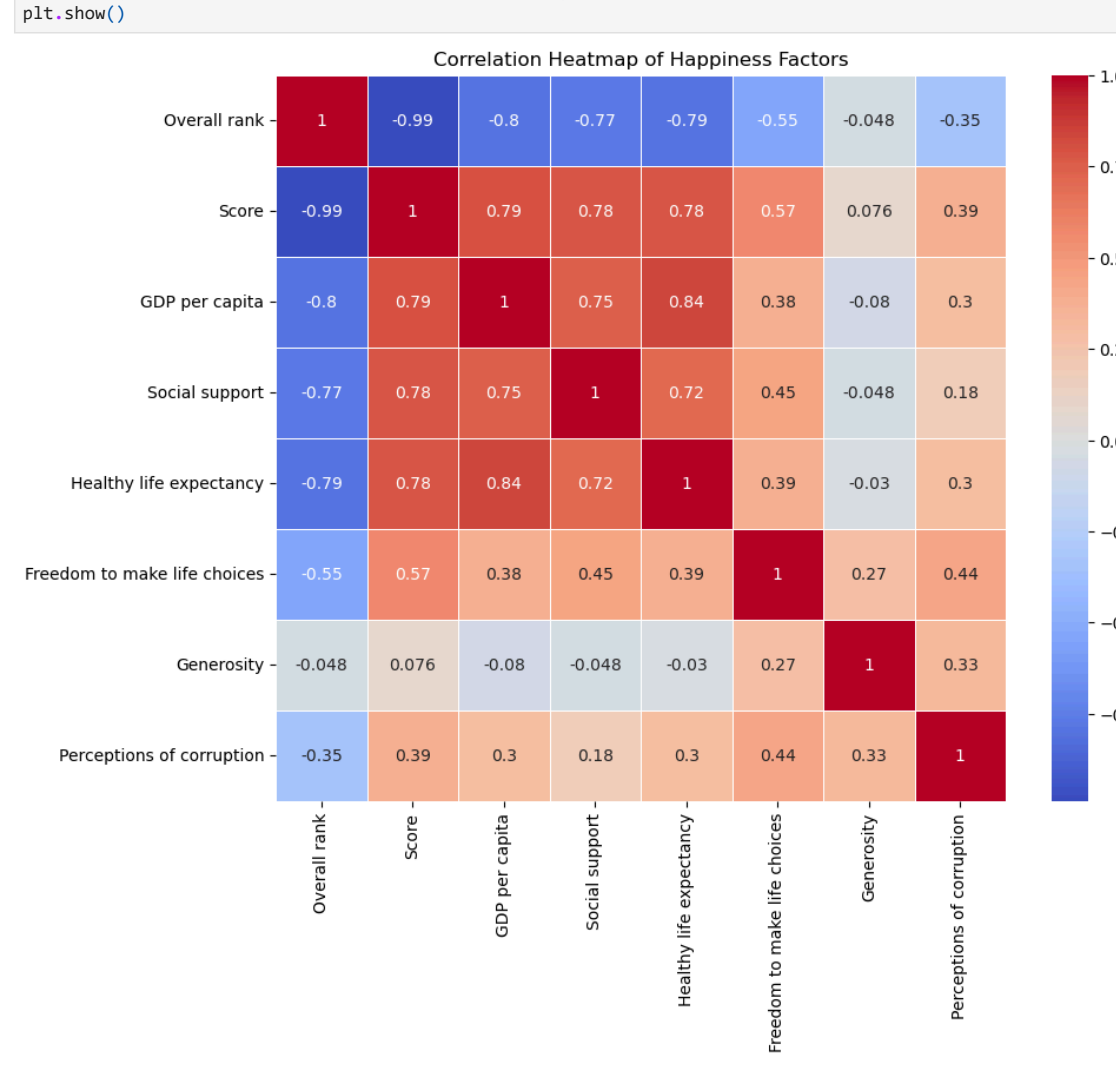


The distribution of happiness scores is right-skewed, as observed in the histogram. This means:

- A majority of countries have happiness scores clustered in the mid-range, between 4 and 6.
- A smaller number of countries have higher happiness scores, closer to 7 or 8.
- The tail of the distribution extends towards the higher end, indicating the presence of outliers or fewer countries with extremely high happiness levels.

Correlation Matrix (Heatmap)

```
In [11]: numeric_columns = df.select_dtypes(include=['float64', 'int64']).columns
         df_numeric = df[numeric_columns]
         correlation = df_numeric.corr()
         plt.figure(figsize=(10, 8))
         sns.heatmap(correlation, annot=True, cmap='coolwarm', linewidths=0.5)
         plt.title('Correlation Heatmap of Happiness Factors')
         plt.show()
```



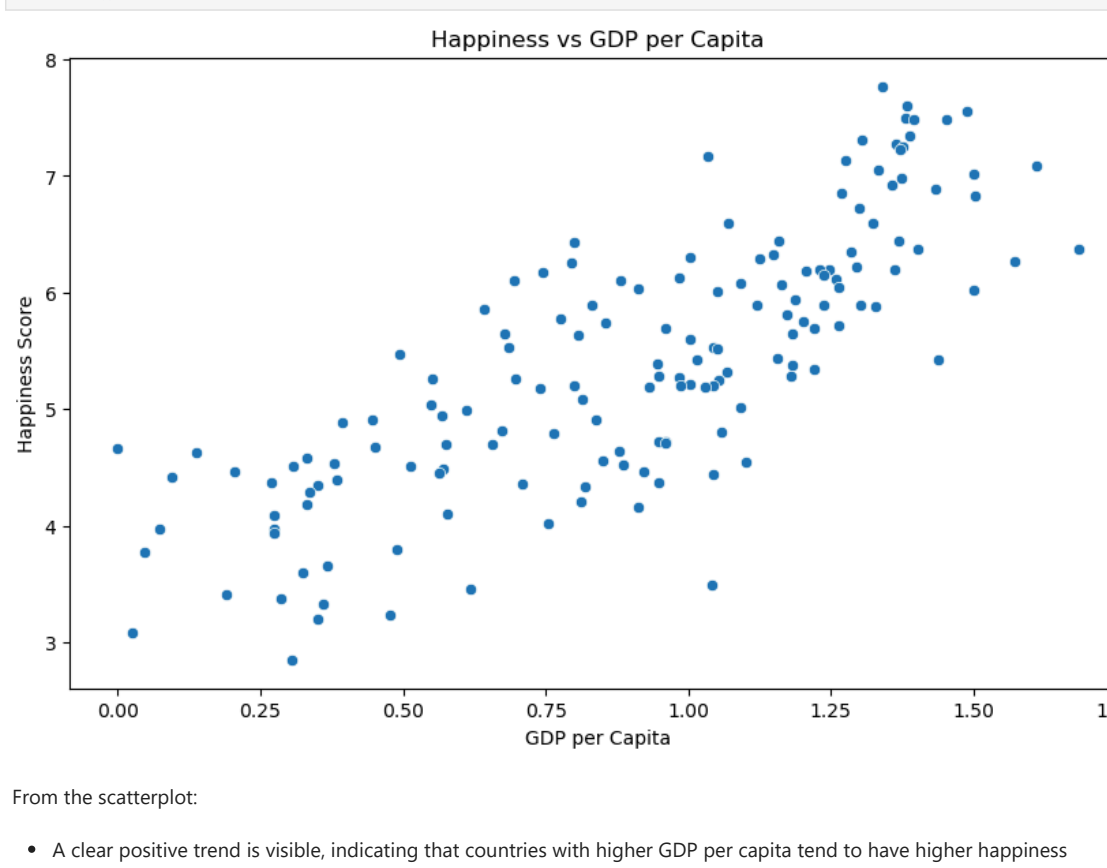
From the heatmap:

- GDP per capita, social support and healthy life expectancy are the strongest contributors to happiness scores, indicating the importance of economic and social stability.
- While generosity and perceptions of corruption also contribute to happiness, their impact are smaller.
- Overall rank is the most strongly related factor since it is based directly on score.

Happiness vs GDP per capita

Examine the relationship between GDP per capita and Happiness score.

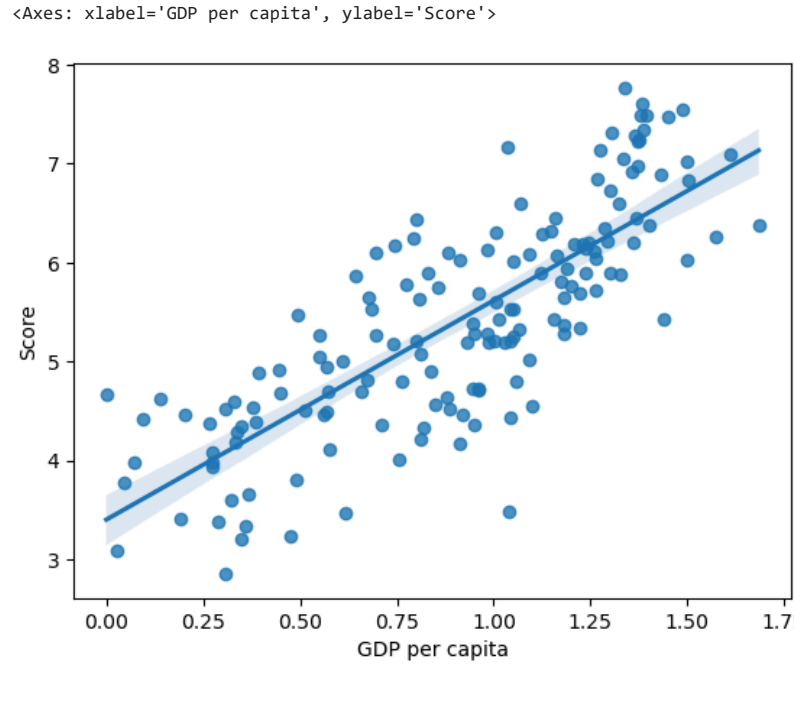
```
In [12]: plt.figure(figsize=(10, 6))
         sns.scatterplot(x='GDP per capita', y='Score', data=df)
         plt.title('Happiness vs GDP per Capita')
         plt.xlabel('GDP per Capita')
         plt.ylabel('Happiness Score')
         plt.show()
```



From the scatterplot:

- A clear positive trend is visible, indicating that countries with higher GDP per capita tend to have higher happiness scores.
- However, the relationship is not strictly linear. For lower GDP values, happiness scores vary widely, suggesting that other factors also play significant roles; For higher GDP values, the spread in happiness scores narrows, implying diminishing returns of wealth on happiness.

```
In [13]: sns.regplot(x='GDP per capita', y='Score', data=df)

Out[13]: <Axes: xlabel='GDP per capita', ylabel='Score'>
```

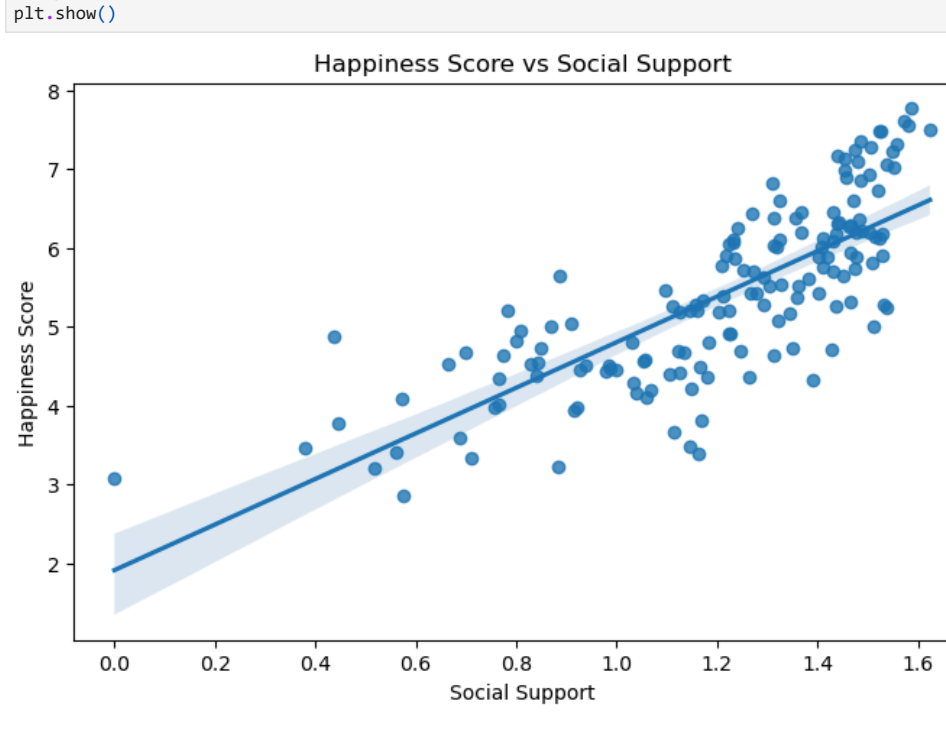


From regression plot:

- The regression line shows a strong positive linear relationship between GDP per capita and Happiness Score.
- The line highlights that GDP per capita can be a reliable predictor of happiness.
- The confidence interval is shown as the shaded area around the regression line, reinforcing that the observed relationship is statistically significant.
- Some points deviate significantly from the regression line, indicating that there are outliers where GDP per capita does not align with the expected happiness levels. Such countries may be looked into further.

Happiness vs Social support

```
In [14]: plt.figure(figsize=(8, 5))
         sns.regplot(x='Social support', y='Score', data=df)
         plt.title('Happiness Score vs Social Support')
         plt.xlabel('Social Support')
         plt.ylabel('Happiness Score')
         plt.show()
```

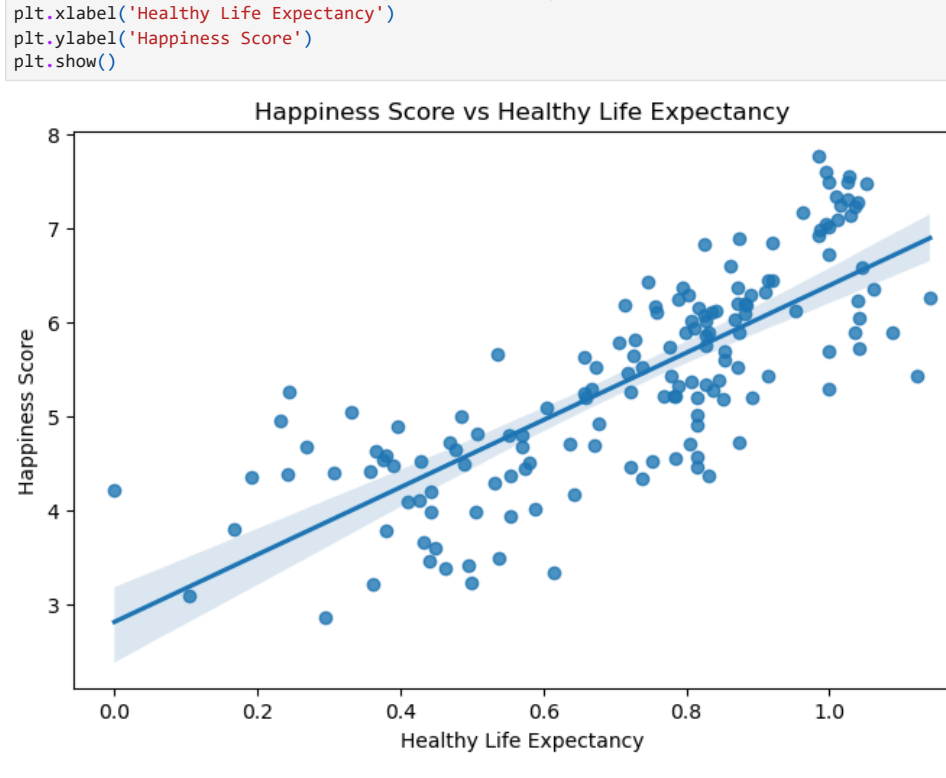


From the regression plot:

- There is a clear positive linear relationship between social support and happiness score. As Social support increases, the happiness score also tends to increase.
- The scatter points closely follow the regression line, indicating a strong correalation, which means social support is likely a significant factor contributing to happiness.
- At higher social support (values>1.2) happiness scores are consistently high; At lower social support (values<1.0), happiness scores vary widely, indicating other factors may influence happiness.

Happiness vs Life expectancy

```
In [15]: plt.figure(figsize=(8, 5))
         sns.regplot(x='Healthy life expectancy', y='Score', data=df)
         plt.title('Happiness Score vs Healthy Life Expectancy')
         plt.xlabel('Healthy Life Expectancy')
         plt.ylabel('Happiness Score')
         plt.show()
```



From the regression plot:

- There is a clear positive correlation between Healthy life expectancy and happiness scroe. As Healthy life expectancy increases, happiness score tends to increase as well.
- The shaded area around the trend line indicates the confidence interval, meaning there is some uncertainty in the exact fit of the line, but the general trend is clear. This would seem to suggest that while health is a major factor, other contributing factors can influence the exact level of happiness.
- The general trend shows that countries with higher Healthy Life Expectancy tend to report higher Happiness Scores.

Predictive Modeling

1. Select features and target

For predicting happiness scores, I am selecting feature variables like GDP per capita, social support, and healthy life expectancy because these could potentially be related to happiness. This is important because in a relevant feature selection, the model will use meaningful information to make predictions.

2. Standardise features

Standardisation adjusts the features so that they have a mean of 0 and a standard deviation of 1. Standardising improves model performance because some machine learning algorithms, including linear regression, are sensitive to the scale of the input features. Features like GDP and healthy life expectancy could have very different scales, for instance, GDP is likely much larger than life expectancy, and standardizing them ensures that no feature dominates due to its scale.

3. Split the data

The splitting of data into training and testing sets means train the model on one part of the data (training set) and then see how it generalizes to new, unseen data (testing set). This step is crucial to prevent overfitting. If only use one set of data for both training and testing, the model might have specific patterns in the data too well and not perform well on new data.

4. Train the model

The model is trained using training data, meaning it learns the relationship between the input features. This step is the core of machine learning where the model will learn from the data through patterns. Training the model allows it to understand how each feature impacts the target and create the coefficients of their relationships.

5. Make predictions

Once trained, the model may be used to predict the target variable (happiness score) for test data. Making predictions on the test to see how well the model applies to new data.

6. Evaluate the model

To evaluate the model use mean squared error (MSE) and R-squared (R^2) to assess how well the model's predictions match the actual values.

MSE

Measure the average squared difference between predicted and actual values. A lower MSE indicates better performance.

R^2

Indicate how well the model explains the variability in the target. A value of 1 mean s the model explains all the varianve while 0 means it does not explain any of the variance.

7. Interpret coefficients

The model coefficients represent the strength and direction of the relationship between each feature and the target variable. The intercept is the predicted value of the target when all features are 0.

Summary

These procedures are required to ensure that your predictive model is well-built, properly assesses its performance, and gives useful insights. Standardising data enhances model performance, splitting data helps prevent overfitting, training the model allows it to learn, and evaluating the model guarantees it can make correct predictions. Interpreting the coefficients enables you to understand the relationships between the features and the target, resulting in actionable insights.

```
In [16]: # 1. Select features and target
         features = df[['GDP per capita', 'Social support', 'Healthy life expectancy']]
         target = df['Score']

         # 2. Standardise features
         scaler = StandardScaler()
         features_scaled = scaler.fit_transform(features)

         # 3. Split the data
         X_train, X_test, y_train, y_test = train_test_split(features_scaled, target, test_size=0.2, random_state=42)

         # 4. Train the model
         model = LinearRegression()
         model.fit(X_train, y_train)

         # 5. Make predictions
         y_pred = model.predict(X_test)

         # 6. Evaluate the model
         print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
         print("R^2:", r2_score(y_test, y_pred))

         # 7. Interpret coefficients
         print("Model Coefficients:", model.coef_)
         print("Intercept:", model.intercept_)

         Mean Squared Error: 0.4636876972797092
         R^2: 0.554456344098441
         Model Coefficients: [0.30476644 0.45051377 0.28976451]
         Intercept: 5.41800352815809
```

MSE

0.4637 indicates that, on average, the squared errors differences between predictions and actual happiness score are small. This means the model is performing reasonably well.

R^2

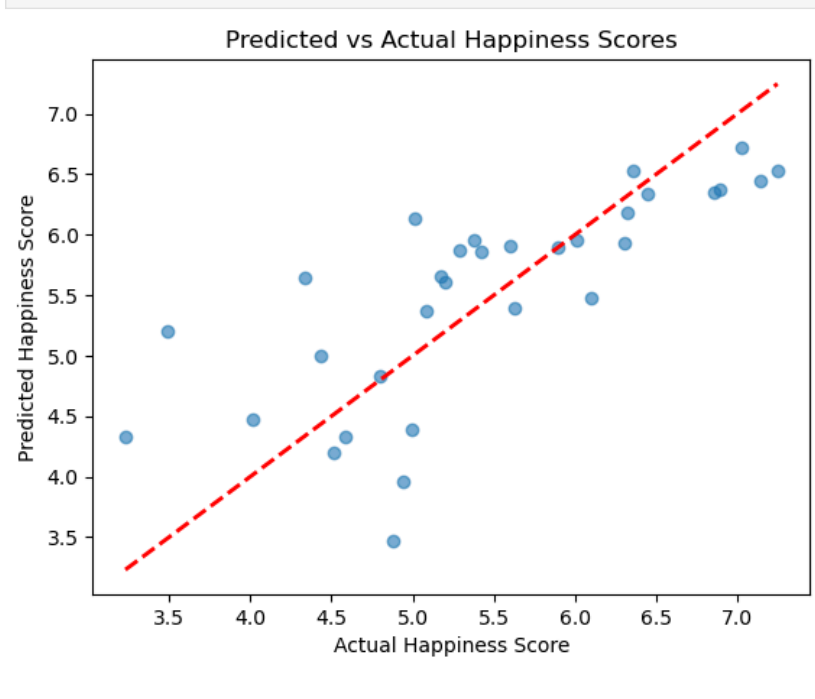
0.5545 means the model explains 55.45% of variability in happiness score based on the features provided. the 45% left unexplained which means other features may also contribute to happiness but not included in the model.

Model coefficient

- GDP per capita: For every 1 unit increase in GDP per capita, the happiness score increases bu approximately 0.3048 points.
- Social support: For every 1 unit increase in social support, the happiness score increases by approximately 0.4505 points.
- Healthy life expectancy: For every 1 unit increase in healthy life expectancy, the happiness score increase by approximately 0.2898 points. #### Intercept: When all features are at their baseline (0 in standardised form), the Happiness Score is expected to be 5.4180. This is the starting point of the model before the features have any effect.

Predicted vs Actual Happiness Score

```
In [17]: plt.scatter(y_test, y_pred, alpha=0.6)
         plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='red', linestyle='--', linewidth=2)
         plt.xlabel('Actual Happiness Score')
         plt.ylabel('Predicted Happiness Score')
         plt.title('Predicted vs Actual Happiness Scores')
         plt.show()
```



The scatterplot compares the actual happiness score (x-axis) with the predicted score (y-axis). The red dashed line represents perfect predictions. The purposes of the graph:

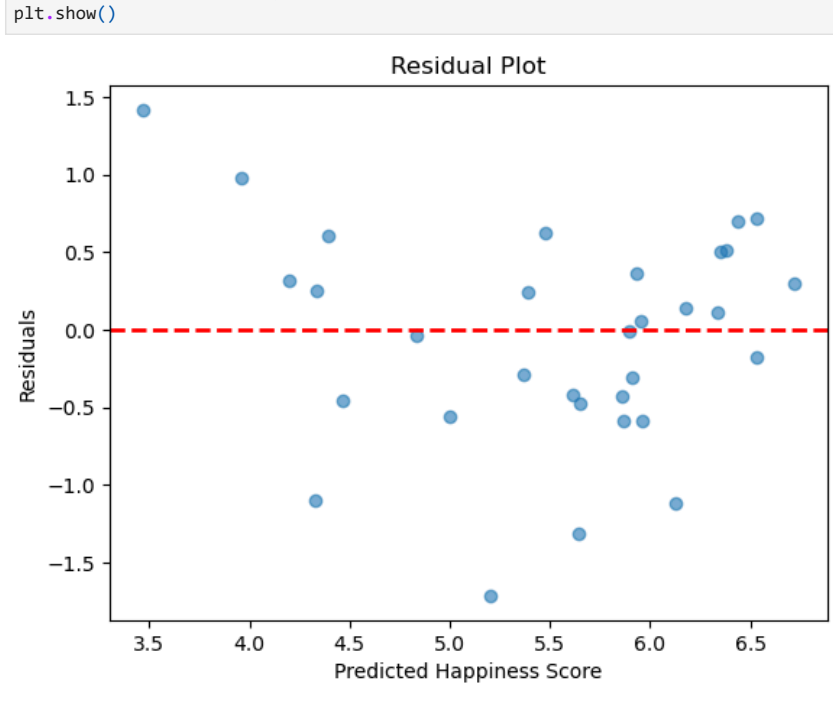
1. Identify how well the model performs overall.
2. Pinpoint areas where the model may underpredict or overpredict.
3. Highlight systematic patterns or errors in the prediction.

From the scatterplot:

- There is a clear positive trend, indicating that the predicted happiness scores generally increase with the actual happiness score.
- Many points lie close to the red dashed line, showing that the predictions are fairly accurate.
- Predictions are more accurate for mid-to-high happiness score.

Residual Plot

```
In [18]: # Calculate residuals
         residuals = y_test - y_pred

         # Residual plot
         plt.scatter(y_pred, residuals, alpha=0.6)
         plt.axhline(0, color='red', linestyle='--', linewidth=2)
         plt.xlabel('Predicted Happiness Score')
         plt.ylabel('Residuals')
         plt.title('Residual Plot')
         plt.show()
```



The residual plot compares the predicted happiness score (x-axis) to residuals (y-axis), which are the differences between actual and predicted values. The red dashed line represents a residual of zero, indicating no error. The purposes of the graph:

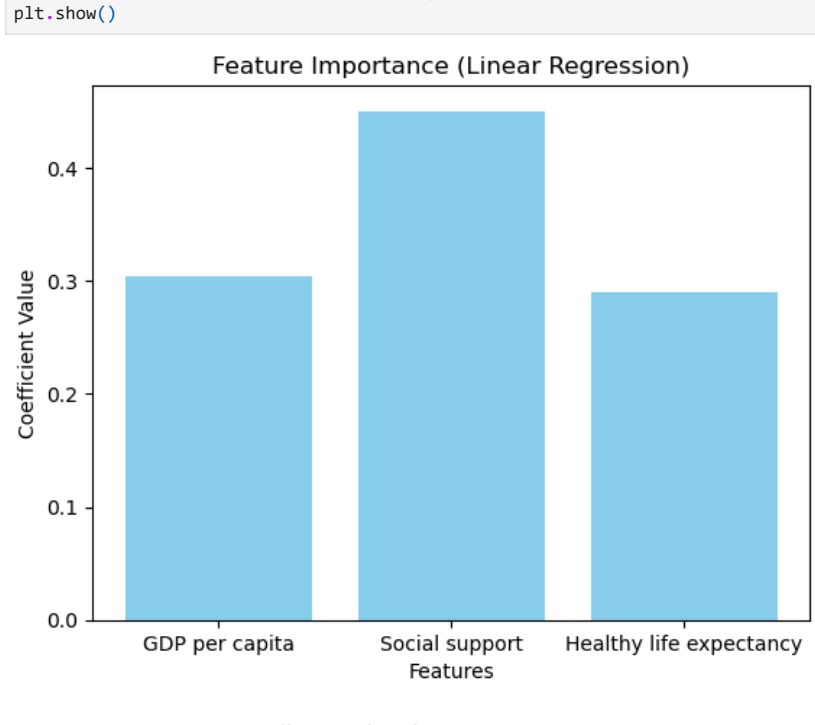
1. To assess the model's performance and detect any patterns in prediction errors.
2. To check for random distribution of residuals, ensuring assumptions of linear regression hold.

From the residual plot:

- The residuals are randomly scattered around the red dashed line, which indicates that the model captures the linear relationship well.
- No clear pattern is visible, which means the model does not systematically underpredict or overpredict values.
- The spread of residuals is fairly constant across predicted happiness score, which reassures that variance is fairly constant.
- A few residuals are further from 0, which may indicate outliers where the predictions are less accurate.

Feature Importance

```
In [19]: # Get feature names and coefficients
         features_names = ['GDP per capita', 'Social support', 'Healthy life expectancy']
         coefficients = model.coef_

         # Bar plot of coefficients
         plt.bar(features_names, coefficients, color='skyblue')
         plt.xlabel('Features')
         plt.ylabel('Coefficient Value')
         plt.title('Feature Importance (Linear Regression)')
         plt.show()
```

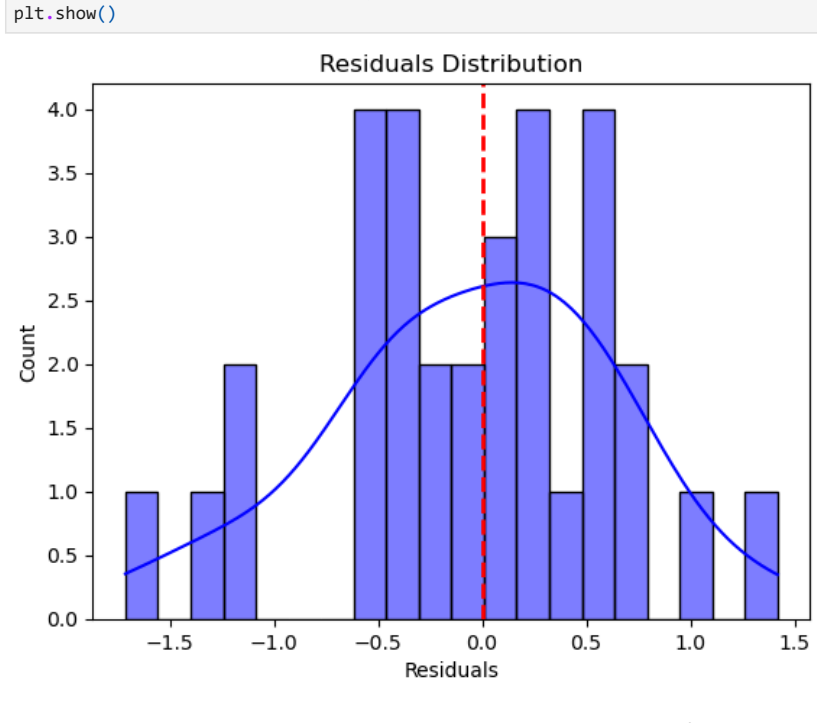


The bar chart displays the coefficients of the features used in the linear regression model. This plot is used to understand which feature contributes most to the prediction of the happiness score. From the bar chart:

- Social support has the highest coefficient, which indicates that it is the most influencing factor for predicting happiness.
- Both the GDP per capita and healthy life expectancy have similar, moderate influences on the score for happiness.
- Positive coefficientss therefore imply that the higher these features, the higher the predicted happiness score is.

Residuals distrubution

```
In [20]: import seaborn as sns

         # Distribution of residuals
         sns.histplot(residuals, kde=True, bins=20, color='blue')
         plt.axvline(0, color='red', linestyle='--', linewidth=2)
         plt.xlabel('Residuals')
         plt.title('Residuals Distribution')
         plt.show()
```



The histogram with a kernel density estimate (KDE) shows the distribution of residuals (errors) from the model predictions. From the histogram:

- The residuals are centered around 0, which means no significant bias in the predictions.
- The distribution is approximately symmetric and resembles a normal distribution. This will satisfy the asumptions of linear regression.
- The red dashed line at 0 represents perfect predictions. Most residuals are close to this line, suggesting the model performs well overall, but some residuals are further out, indicating minor prediction errors.

Discussion of results

Features Importance

The analysis confirms that social support is the strongest predictor of happiness, followed by GDP per capita and healthy life expectancy.

Model Accuracy

While the model performs well, explaining 55.45% of the variability in happiness scores ($R^2 = 0.5545$), some variance remains unexplained. This indicates the presence of other unmeasured factors influencing happiness.

Limitations:

- Dataset bias: Over-representation of wealthier countries.
- Excluded factors: Political stability, personal freedoms, and environmental quality are not included, which could impact happiness scores.

Conclusion

This project has successfully demonstrated the relationships between happiness score and socio-economic indicators such as GDP per capita, social support and healthy life expectancy. The predictive model reinforced the importance of these factors as key determinants of global well-being. However, the findings also emphasise that happiness is multifaceted and non-economic factors are very important. Other indicators of environmental quality, political freedoms and mental health data can be included in future research to improve the model. These findings can be used by policymakers to develop more specific interventions aimed at economic and social welfare.

Recommendations for future research

- Including data on environmental quality, political stability and mental health could improve the model's accuracy.
- A deeper analysis of low-income countries could address the dataset's bias.
- Subjective factors like individual well-being and life satisfaction could be integrated into the analysis.

References

Wikipedia (2024) Happiness. Online. Available at: https://en.wikipedia.org/wiki/Happiness [Accessed: 4 December 2024]. </br> Sachdeva, P. (2021) Should investors consider happiness rather than GDP? Online. Available at: https://www.schroders.com/en-gb/uk/intermediary/insights/should-investors-consider-happiness-rather-than-gdp/ [Accessed: 4 December 2024]. </br> Kaggle (2019) World Happiness Report. Online. Available at: https://www.kaggle.com/datasets/unsdsn/world-happiness [Accessed: 6 December 2024]. </br> The Pecan Team (2024) Predictive modeling. Available at: https://www.pecan.ai/blog/predictive-modeling/ [Accessed: 9 December 2024]. </br> GeeksforGeeks (2024) Residual Plots for Nonlinear Regression. Available at: https://www.geeksforgeeks.org/residual-plots-for-nonlinear-regression/ [Accessed 18 July 2024]. </br>

```
In [21]: !pip list > listOFpackages.txt
```

```
In [22]: input_file = "listOFpackages.txt"
         output_file = "requirements.txt"

         with open(input_file, "r") as infile, open(output_file, "w") as outfile:
             for line in infile:
                 if "Package" in line or "-" in line:
                     continue  # Skip header and separator lines
                 parts = line.split()
                 if len(parts) == 2:  # Ensure it's a valid package line
                     package, version = parts
                     outfile.write(f"{package}=={version}\n")
```

```
In [ ]:
```